How to teach sign language to a chimpanzee

In the 1960s and '70s, a group of chimpanzees astonished the world by learning sign language. Only two remain, and one question still lingers: Was it worth it?

To psychologists in the behaviorist school, which arose in the early 1900s, animal behavior was merely a response to external stimuli. They believed that other species' inability to talk wasn't just evidence they lacked the proper anatomical structures; it made studying their inner life, a concept the school found largely inaccessible, impossible. For some, it was even evidence of a lack of true consciousness. Linguists, including Noam Chomsky, pushed the belief that humans alone had an ingrained natural grammar, from which all languages emerged. As the thinking went, animals would have no use for language.

The Gardners, however, believed that chimps would have plenty to say if given the proper tools to do so.

They weren't the first to suspect chimps were capable of communication. In the 1930s another pair of married scientists took a chimpanzee named Gua into their home to see what would happen if they raised Gua in the same manner as a human child. That included teaching Gua English while they raised him alongside their human son, Donald. As documented in their book *The Ape and the Child,* Winthrop and Luella Kellogg found that the chimp was actually faster at learning to use a spoon and open a door, as well as responding to spoken commands. However, they abandoned the experiment after Donald began speaking and Gua did not. Further experiments conducted beginning in the 1940s showed that chimps lacked the necessary anatomical structures to speak English—though one chimp, named Viki, was ultimately able to say mama, papa, and cup after intensive speech therapy that involved humans manually shaping her mouth to make the right sounds.

With human vocalizations out, the Gardners theorized that apes would be able to learn ASL much like the way they learn to communicate via gestures in the wild. They'd have to be fully immersed, surrounded by human caregivers who would communicate with them and each other solely in ASL.

Tatu and Loulis are the last two surviving chimps from a series of experiments that began in the 1960s. The pair have spent their entire lives in captivity, including stints at facilities in Oklahoma and Washington, where researchers studied their ability to learn ASL—hoping to challenge the prevailing belief that only humans could use language.

Since 2013 they've been living at Fauna, where both chimpanzees continue to sign. Tatu, the more loquacious one, has mastery of 215 signs, while Loulis can use 78. Among their favorite things to talk to their caregivers about are food and activities. Tatu is partial to cheese and enjoys a nice cup of coffee or tea, while Loulis prefers asking his caregivers to play a game of chase.

Whether language is unique to humans has been a subject of fascination to linguists and scientists for centuries, and in the 1960s they began turning in earnest to one of our closest genetic relatives to finally put the question to rest. This research was pioneered by a pair of psychology professors at the

University of Nevada, Reno, the husband-and-wife couple Allen and Beatrix “Trixie” Gardner, who sought to prove that under the right circumstances chimpanzees could learn a human language.

Their research began in 1966 with an infant chimp, taken from West Africa for use in U.S. Air Force experiments, whom they adopted and named Washoe. The Gardners raised Washoe as much like a human child as possible. She learned to use a potty (which she would call “dirty good” in sign language), played with dolls, and admired the photos in magazines.

Washoe became a sensation in academia, but her fame would be surpassed in the 1970s by two other primates: Koko the gorilla and Nim Chimpsky. While the methodologies for those experiments would later come under fire, casting into doubt whether Koko and Nim understood what they were saying, the same was not true for Washoe, who ultimately developed a vocabulary of 245 words before her death in 2007 at Central Washington University.

The Gardners repeated their experiments with four more chimpanzees: Moja, who loved to paint; Dar, who enjoyed wearing hats and examining mechanical objects; Pili, a male who died from leukemia before age two; and Tatu. They eventually turned their work over to their former graduate student, Roger Fouts, and his wife, Deborah, who took in Washoe, Moja, Dar, and Tatu, as well as young Loulis, who became the first and only chimp to learn ASL from other chimps.